

Statement of participation

Oluwaseun Ajayi

has completed the free course including any mandatory tests for:

What chemical compounds might be present in drinking water?

This 6-hour free course examined the chemical compounds that may occur in drinking water and their potential effects.

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www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:

<https://www.open.edu/openlearn/science-maths-technology/what-chemical-compounds-might-be-present-drinking-water/content-section-0>

COURSE CODE: S215_1

What chemical compounds might be present in drinking water?

<https://www.open.edu/openlearn/science-maths-technology/what-chemical-compounds-might-be-present-drinking-water/content-section-0>

Course summary

Do you want to know what's in the water that you drink? This free course, What chemical compounds might be present in drinking water?, examines the chemical compounds that occur in drinking water. A high level of certain anions in water can cause environmental pollution and health problems. Cations are also important. For example, calcium salts contribute to the hardness of water and water treatment uses aluminium compounds. Additionally oxidising agents, such as ozone or chlorine, are vital in the disinfection of water. This OpenLearn course focuses on the chemistry of the p-block elements in Groups 13 to 18 of the Periodic Table.

Learning outcomes

By completing this course, the learner should be able to:

- understand that in drinking water anions have a beneficial range of concentration above which they may have an adverse effect on either human health or the environment. Toxic elements and compounds often block essential biological processes
- explain how excess anions or cations in aqueous solution can be removed by ion-exchange chromatography or an ion-exchange membrane
- explain how cations such as aluminium can be amphoteric, that is they can neutralise both acidic and alkaline solutions. Also recognise how this enables aluminium(III) ions to be used in water treatment.

Completed study

The learner has completed the following:

Section 1

Water

Section 2

Anions in water

Section 3

Phosphorus compounds in water

Section 4

Cations in water